

Caleb Erickson

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Education

Master of Business Administration, Cybersecurity Focus May 2027
The University of Alabama, Manderson Graduate School of Business, Tuscaloosa, AL

- GPA: 4.0/4.0
- Combined MS/BS Program

Bachelor of Science in Computer Science May 2026
The University of Alabama, Lee J. Styslinger Jr. College of Engineering, Tuscaloosa, AL

- GPA: 4.0/4.0
- Most Outstanding Undergraduate Award
- Honors College
- President's List

Skills

- **Languages:** Python, C++, C, JavaScript, TypeScript
- **Databases:** PostgreSQL, MongoDB, Azure Cosmos DB, Neo4j, Milvus
- **Tools:** Git, AWS, Azure, Google Cloud, Docker
- **Frameworks:** REST APIs, Django, Flask, React, Next, Node
- **Data Libraries:** NumPy, Pandas, Matplotlib, Scikit-learn

Experience

The Alabama Institute for AI April 2025 - Present
Artificial Intelligence & Machine Learning Intern

- Build an automated pipeline integrating 5 federal datasets into a 50K-node knowledge graph
- Develop a RAG framework integrating vector embeddings, LLMs, and knowledge graphs, reducing hallucinations by 38% through reranking, improved embedding quality, and optimized document chunking
- Lead development of a full-stack web app used in 6 university courses, supporting ~75 users for AI data storytelling
- Present and demo AI/ML systems at NSF OKN Year 3 Kickoff in Washington, D.C. to an audience of ~250 attendees

QuantHub August 2025 - April 2026
Software Development Intern

- Built a scalable web scraper extracting enrollment data from 41,467 course sections across 10, normalizing 14,083 course records from 153 subjects into a cloud-hosted SQL database

Projects

Mockstreet Stock Exchange, *mockstreetexchange.com* March 2025

- Engineered a mock stock exchange platform enabling users to simulate trades and manage portfolios using real-time market data via the Yahoo Finance API; scaled to ~100 concurrent users

Number Prediction Neural Network, *numbers-beige.vercel.app* April 2025

- Designed and trained a custom convolutional neural network for handwritten digit recognition, achieving 96.4% accuracy on MNIST dataset with a browser-based interactive interface

Portfolio with All Projects, *caleberickson21.github.io*

Publications

- **Erickson, C., et al.** *Enhancing Flood Risk Communication through Semi-Automated Ontology Framework*. *Water*, 2025. doi:10.3390/w17192801 August 2025
- **Erickson, C., et al.** *FLAI: An AI Framework for Flood Risk Communication Using Knowledge Graphs and RAG*. *Hydrology*, 2025. doi:10.3390/hydrology12080204 August 2025

Involvement

- Alabama AI Club, *Developer and Executive Officer* August 2024 - Present
- Alabama Software Development Club, *Member* August 2023 - May 2025